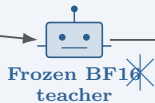


Stage I: Quantization-aware self-distillation

Prompts

$$x \sim \mathcal{D}$$



y , top- k logits

$$\mathcal{L}_{\text{KD}} = T^2 \sum_{t \in y} \text{KL}(p_T \| p_S)$$



Stage II: FP4 GRPO post-training

QAT-adapted
FP4 policy π_θ

Quantized rollouts
 $\{y_i\}_{i=1}^G$

Task rewards and
group advantages

Update policy
with GRPO

FP4 linear operator inside the policy

